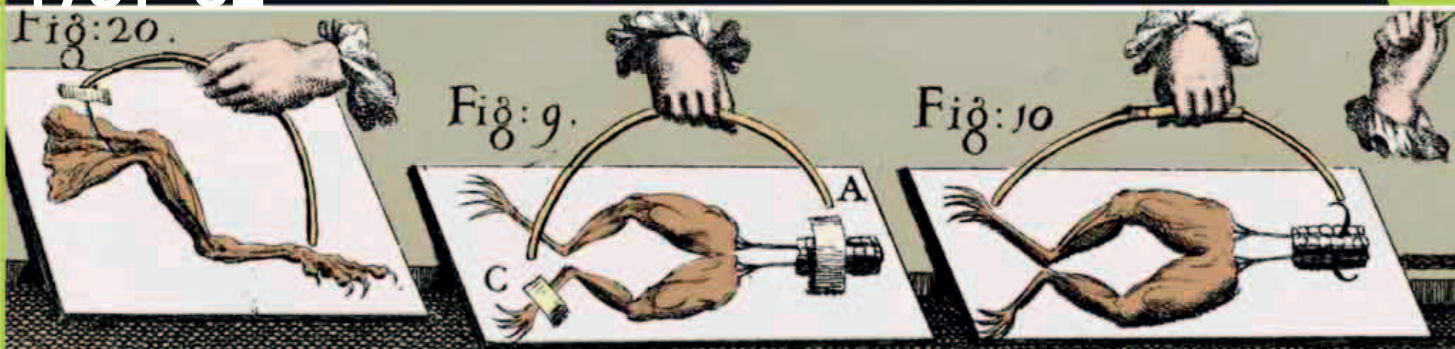


418

TONS

THE **AMOUNT**
OF **IRON** USED
TO MAKE
THE **BRIDGE**

1781–82



Macabre by today's standards, Galvani's experiments in generating movement in the leg muscles of dead frogs were a key step in the understanding of electricity.



JAN INGENHOUSZ
(1730–99)

Born in Breda in the Netherlands, Ingenhousz was a physician who had wide-ranging interests in other scientific fields, including the study of electricity. He was one of the pioneers of vaccination against smallpox and successfully inoculated the Austrian Empress Maria Theresa and her family.

previously held theory that body heat comes from phlogiston. In fact, animals make heat by combustion using oxygen.

Swiss physicist **Almé Argand** (1750–1803) revolutionized home lighting with the **Argand oil lamp**, 10 times brighter than a candle. The lamp had a circular wick and a tall glass chimney to improve the airflow.

THE SCIENTIFIC SENSATION OF 1781 WAS THE DISCOVERY OF A SIXTH PLANET

by German-born British astronomer **William Herschel** (1738–1822). When Herschel first saw the object through the telescope in his garden in Bath in England on March 13, he assumed it was a comet. But its brightness and elliptical orbit soon left astronomers in no doubt. Herschel named the new planet George's Star in honor of British King George III, but astronomers eventually decided on **Uranus**, after Ouranos, the Greek god of the sky.

French astronomer **Charles Messier** (1730–1817) published the final version of his *Catalogue des Nébuleuses et des Amas d'Étoiles* (**Catalog of Nebulae and Star Clusters**) in 1781. This list, which Messier had been compiling since the 1770s, recorded 103 vague, distant objects in the night sky, all still known to modern astronomers by the numbers Messier gave them. Also in 1781, another astronomer, the Czech **Christian Mayer** (1719–83), published a catalog listing 80 pairs

Messier's objects

Many of the objects in Messier's list have been revealed to be distant galaxies far from our own Milky Way, which in Messier's time was thought to comprise the entire Universe.

1,788

MILLION MILES

Uranus

93

MILLION MILES

Earth

Distance from the Sun

Uranus orbits the Sun at a distance of 1,788 million miles, almost 20 times the distance of Earth from the Sun. Each orbit takes 84 Earth years.

of stars, later known as **binaries**, that John Michell had argued in 1767 revolve together and which were observed by astronomers such as Herschel.

The same year, Italian physician **Luigi Galvani** (1737–98) began

carrying out experiments showing that **static electricity** could cause the muscles of dissected frogs' legs to twitch. In April, he connected the nerves of a dead frog to a metal wire pointed skyward in a thunderstorm. In this experiment, which inspired Mary Shelley's novel *Frankenstein*, the frog's legs jumped with each flash of lightning. In September, the legs twitched when just hung on a brass hook over a metal rail. A fierce argument sprang up between Galvani and fellow Italian experimenter **Alessandro Volta** over whether the electrical effect was intrinsic to muscles and life (animal electricity), as Galvani claimed, or whether it was a chemical reaction, as Volta believed. Animal and chemical electricity were later shown to be the same.

In 1782, Lavoisier formulated what is perhaps now regarded as the most **important law in chemistry**—that of the conservation of matter. This law shows that in any chemical reaction no matter—not even the smallest part—is ever lost; it is simply rearranged in different combinations.

103

THE NUMBER OF
ASTRONOMICAL
OBJECTS IN
MESSIER'S LIST



1780 Lavoisier shows animals produce heat by combustion

January 26, 1781 Luigi Galvani begins experiments with static electricity

1781 Charles Messier publishes his list of nebulae and star clusters

March 13, 1781 William Herschel discovers the planet Uranus

1782 Lavoisier formulates the law of the conservation of matter

1780 Almé Argand invents the Argand oil lamp

1781 Christian Mayer publishes a catalog of 80 binary stars

April 26, 1781 Galvani shows how lightning activates the muscles of a dead frog